

Feature Store Documentation

RecoMart Recommendation Pipeline | Task 7 Deliverable

1. Overview

The RecoMart pipeline utilizes Feast as a centralized Feature Store to serve features consistently for both model training (offline) and real-time inference (online). The offline store relies on Parquet files, while the online store is powered by SQLite.

2. Feature Metadata Documentation

Entities: The primary entities are users and products, mapped via user_id and product_id.

2.1 Product Feature View

Name: product_features | Source: data/features/product_features.parquet

Feature Name	Data Type	Description
click_to_purchase_ratio	Float32	Ratio of clicks leading to a purchase
price_bracket	Int32	Price quartile (0=Low, 3=Premium)
brand_popularity	Float32	Normalized demand score for the brand
quality_score	Float32	Weighted product rating and sentiment
rating	Float32	Product rating score (0.0 to 5.0)
price	Float32	Product price in INR
sentiment_score	Float32	Customer review sentiment (-1.0 to 1.0)

2.2 User Feature View

Name: user_features | Source: data/features/user_features.parquet

Feature Name	Data Type	Description
activity_frequency	Int32	Total number of interactions by the user
avg_rating_given	Float32	Mean rating submitted by the user
rating_variance	Float32	Variance in user ratings (consistency)
unique_items_interacted	Int32	Count of unique products interacted with
purchase_ratio	Float32	Proportion of interactions that are purchases

3. Sample Feature Retrieval Demonstration

Features can be retrieved from the online store for real-time inference using the Feast Python SDK.

```
from feast import FeatureStore

# Initialize the Feature Store
store = FeatureStore(repo_path="data/feast")
```

```
# Retrieve features for a specific user and product
feature_vector = store.get_online_features(
    features=[
        "user_features:activity_frequency",
        "user_features:avg_rating_given",
        "product_features:quality_score",
        "product_features:price_bracket"
    ],
    entity_rows=[{"user_id": 42, "product_id": 1024}]
).to_dict()

print(feature_vector)
```